PSEUDO-COHERENT COMPLEXES ARE HOMOTOPY COLIMITS OF PERFECT COMPLEXES APPROXIMATING ALL TRUNCATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Perfect objects are pseudo-coherent and pseudo-coherence of a complex is detected by the pseudo-coherence of all its truncations. invoked at proof lines 4825-4837. Runs the easy direction (2) implies (1).
- Every pseudo-coherent complex on a quasi-compact quasi-separated scheme admits approximations by perfect complexes up to any amplitude, and maps into the cone of an approximation can be killed by choosing the next approximation deep enough via the Hom vanishing lemma. invoked at proof lines 4840-4862. Constructs the inductive system with the factoring maps.
- A map into a complex whose cone has no Hom from the source factors through any given approximation. invoked at proof lines 4864-4867. Provides the factoring maps K_n to K_{n+1} completing the induction step.

Source reference. Original lemma 08CQ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4827-4829 for the easy direction; the truncation bookkeeping is inside the counted span.

Source reference. Original definition 08CB: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4835-4837 to conclude K is pseudo-coherent; the n-uniform argument is inside the counted span.

Source reference. Original definition 08EM: sources/stacks-perfect/pinned-original-source.tex, source lines 3814–3814. Cited at proof lines 4841-4844 among the approximation definitions; the construction of the system is inside the counted span.

Source reference. Original definition 08EN: sources/stacks-perfect/pinned-original-source.tex, source lines 3841–3841. Cited at proof lines 4841-4844 for the approximations K_1 and K_{n+1}; the induction consuming them is inside the counted span.

Source reference. Original theorem 08ES: sources/stacks-perfect/pinned-original-source.tex, source lines 4031–4031. Cited at proof lines 4842-4845 for the existence of the approximations; the inductive construction is inside the counted span.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 09M4: sources/stacks-perfect/pinned-original-source.tex, source lines 3465–3465. Cited at proof lines 4847-4850 to pick a, b for K_n; the killing of Hom(K_n, C) is inside the counted span.

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4864-4867 to factor K_n to K through K_{n+1}; the induction step is inside the counted span.

Source reference. Original lemma 0CRK: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4869-4872 to conclude K = hocolim K_n; the verification against the constructed system is inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. Let $K \in D(\mathcal{O}_X)$. The following are equivalent*

1. *K is pseudo-coherent, and*
2. *$K = \text{hocolim} K_n$ where K_n is perfect and $\tau_{\geq -n}K_n \to \tau_{\geq -n}K$ is an isomorphism for all n.*

Proof. The implication (2) $\Rightarrow$ (1) is true on any ringed space. Namely, assume (2) holds. Recall that a perfect object of the derived category is pseudo-coherent, see Cohomology, Lemma 08CQ. Then it follows from the definitions that $\tau_{\geq -n}K_n$ is $(-n+1)$-pseudo-coherent and hence $\tau_{\geq -n}K$ is $(-n+1)$-pseudo-coherent, hence K is $(-n+1)$-pseudo-coherent. This is true for all n, hence K is pseudo-coherent, see Cohomology, Definition 08CB.

Assume (1). We start by choosing an approximation $K_1 \to K$ of $(X, K, -2)$ by a perfect complex K_1, see Definitions 08EM and 08EN and Theorem 08ES. Suppose by induction we have

$$K_1 \to K_2 \to \ldots \to K_n \to K$$

with K_i perfect such that such that $\tau_{\geq -i}K_i \to \tau_{\geq -i}K$ is an isomorphism for all $1 \leq i \leq n$. Then we pick $a \leq b$ as in Lemma 09M4 for the perfect object K_n. Choose an approximation $K_{n+1} \to K$ of $(X, K, \min(a-1, -n-1))$. Choose a distinguished triangle

$$K_{n+1} \to K \to C \to K_{n+1}[1]$$

Then we see that $C \in D_{QCoh}(\mathcal{O}_X)$ has $H^i(C) = 0$ for $i \geq a$. Thus by our choice of a, b we see that $\text{Hom}_{D(\mathcal{O}_X)}(K_n, C) = 0$. Hence the composition $K_n \to K \to C$ is zero. Hence by Derived Categories, Lemma 0149 we can factor $K_n \to K$ through K_{n+1} proving the induction step.

We still have to prove that $K = \text{hocolim} K_n$. This follows by an application of Derived Categories, Lemma 0CRK to the functors $H^i(-) : D(\mathcal{O}_X) \to \mathit{Mod}(\mathcal{O}_X)$ and our choice of K_n. $\square$